INCH-POUND
MIL-DTL-83507/11A
25 July 2025
SUPERSEDING
MIL-T-83507/11(AS)
16 March 1978

DETAIL SPECIFICATION

Tool Kit, Electrical Connectors, Contacts, and Connector and Cable Accessories, Tool Kit for SAE-AS22520/1 Crimping Tool and Turrets

Inactive for new design.

This specification is approved for use by all Departments and Agencies of the Department of Defense.

The complete requirements for procuring the tool kit described herein shall consist of this document and the latest issue of Specification MIL-DTL-83507.

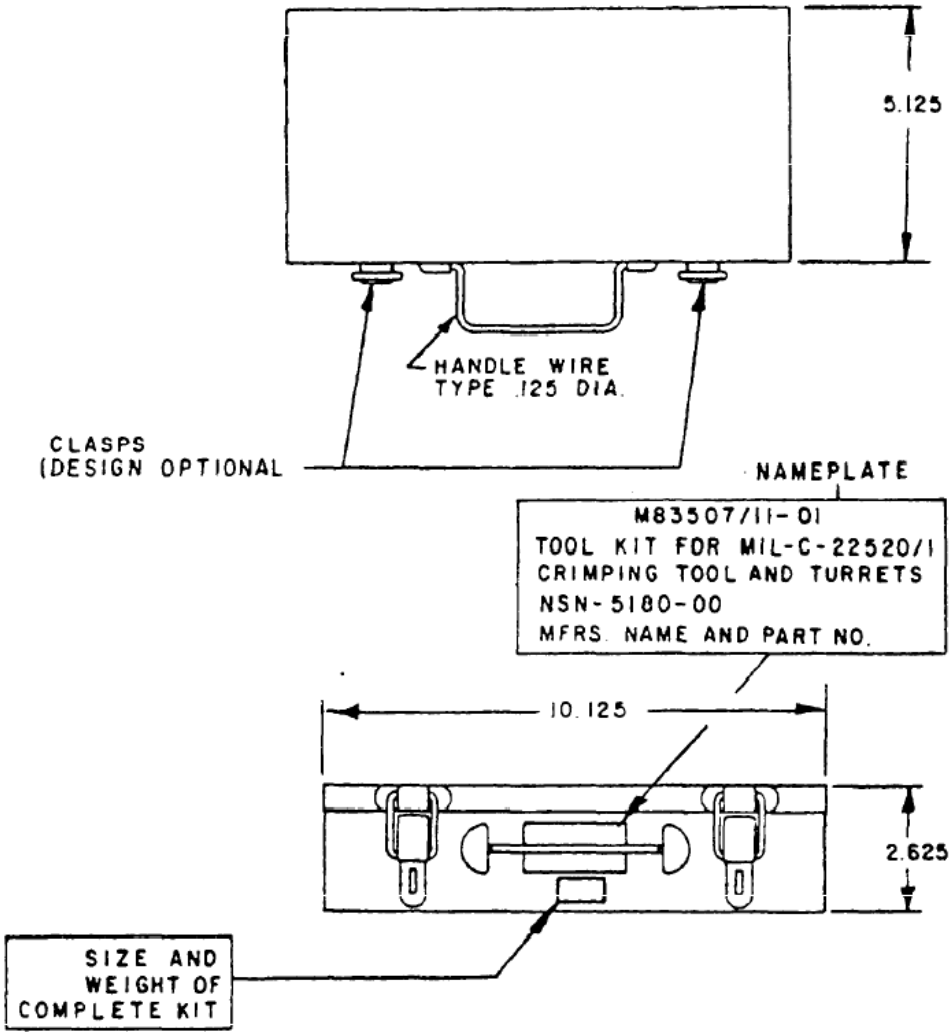


FIGURE 1 Metal case. Tool kit for SAE-AS22520/1 crimping tool and turrets.

DISTRIBUTION STATEMENT A. Approved for public release; distribution is unlimited.

Source: https://assist.dla.mil -- Downloaded: 2026-10-23T08:22Z
Check the source to verify that this is the current version before use.

NOTES:
1. Part number M83507/11-01 includes case and tools.
2. Part number M83507/11-02 for case only (tools not included).

Parts List:

(1) M83507/ll-02	-	Tool case assembly.
(l) M22520/1-01	-	Crimp tool frame.
(l) M22520/1-02	-	Turret head assembly.
(l) M22520/1-03	-	Turret head assembly.
(l) M22520/3-1	-	In-service inspection gage.

(1) 9/64 hex wrench.
(1 set) - Contents chart, instruction sheets and tool selection chart.

FIGURE 1 Metal case. Tool kit for SAE-AS22520/1 crimping tool and turrets. – Continued

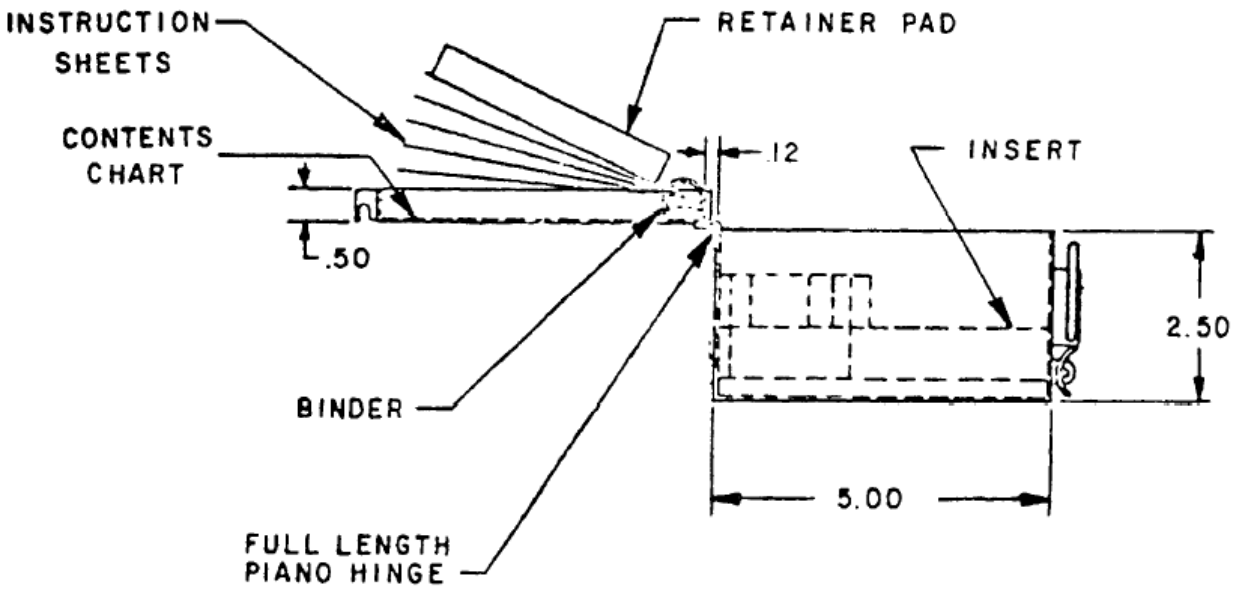


FIGURE 2 Metal case. Tool case assembly.

Source: https://assist.dla.mil -- Downloaded: 2026-10-23T08:22Z
Check the source to verify that this is the current version before use.

Instruction sheets are as follows:

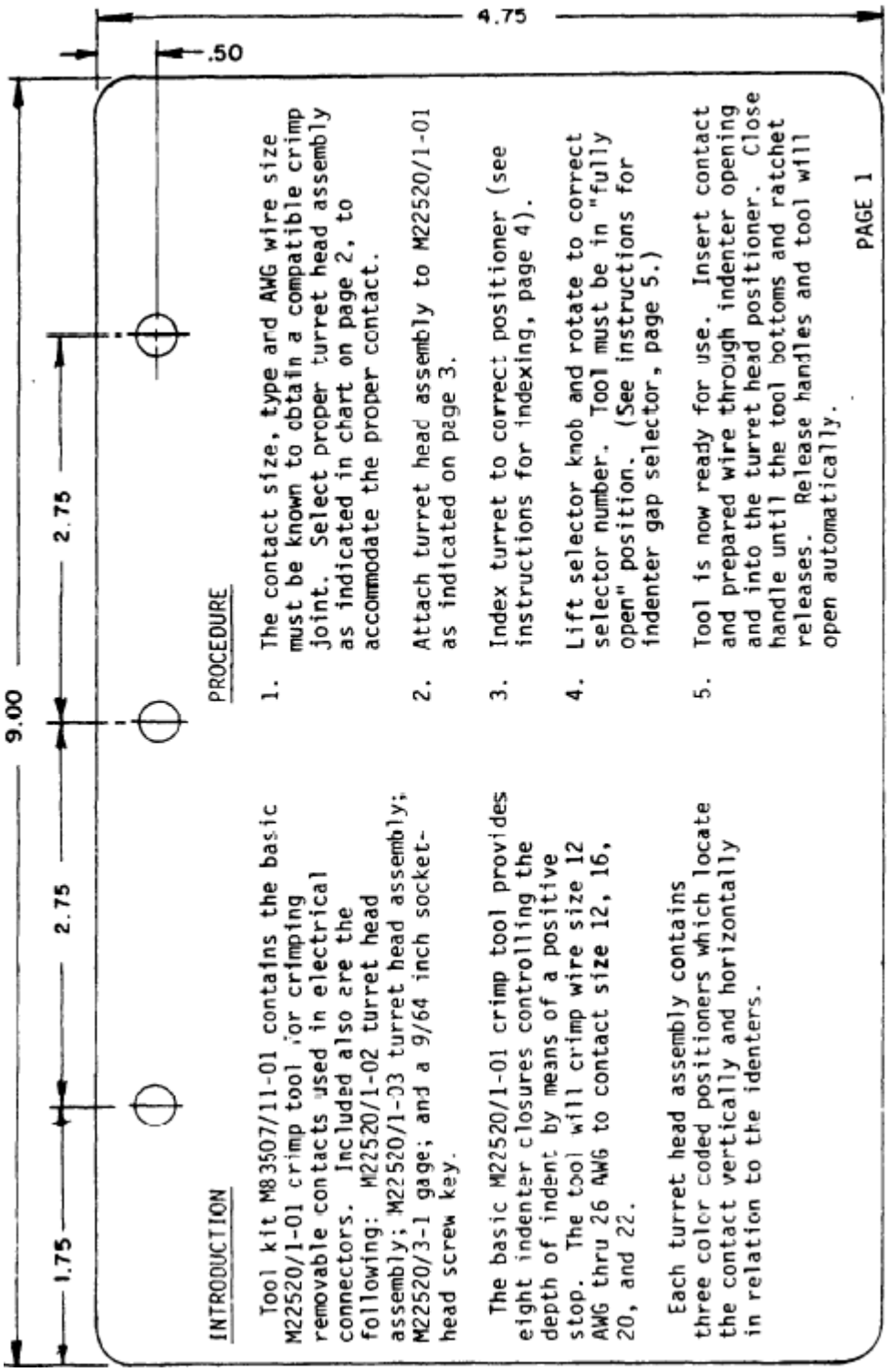


FIGURE *6*

FIGURE 3 Metal case. Insert.

Source: https://assist.dla.mil -- Downloaded: 2026-10-23T08:22Z
Check the source to verify that this is the current version before use.

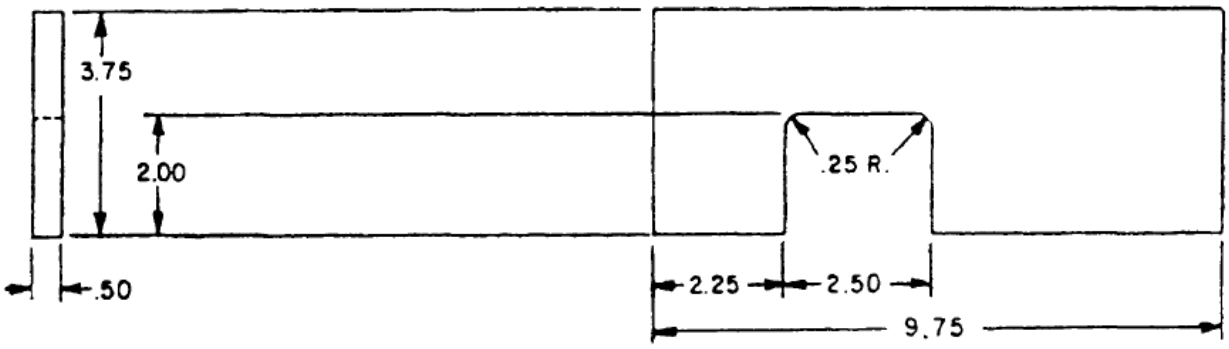


FIGURE 4 Metal case. Retainer.

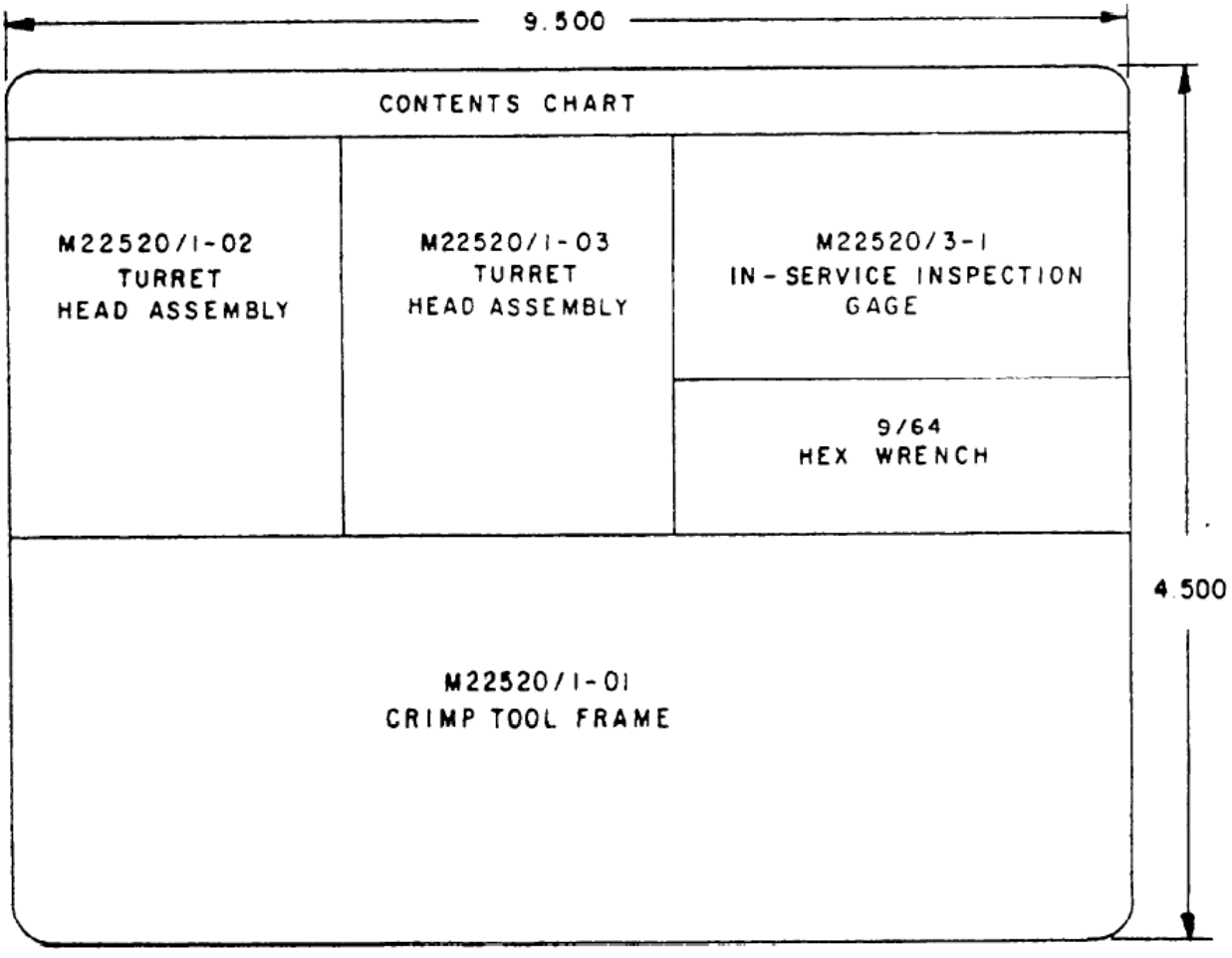


FIGURE 5 Metal case. Contents chart.

Source: https://assist.dla.mil -- Downloaded: 2026-10-23T08:22Z
Check the source to verify that this is the current version before use.

Instruction sheets are as follows:

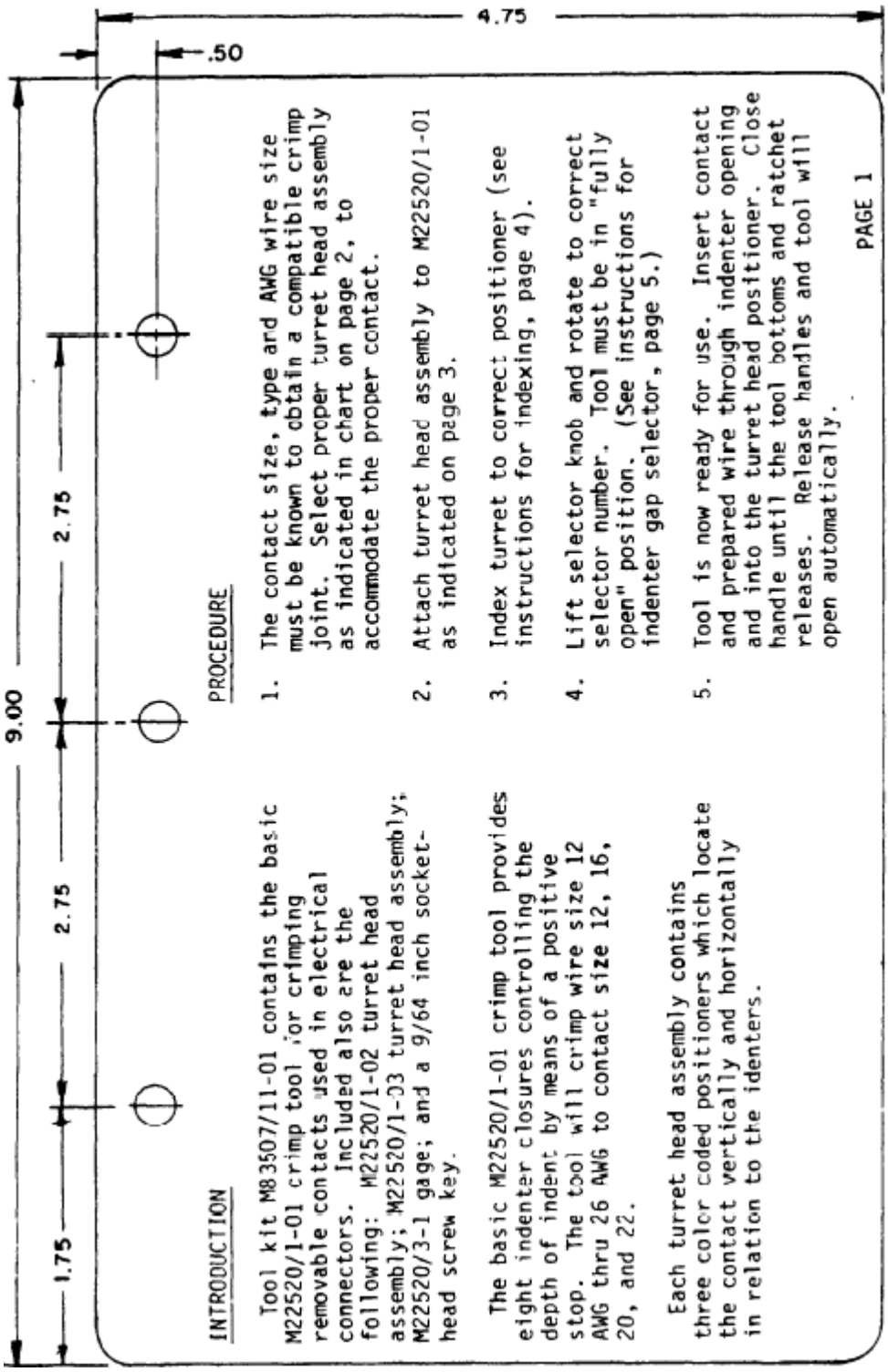


FIGURE 6 Metal case. Instruction sheet.

Source: https://assist.dla.mil -- Downloaded: 2026-10-23T08:22Z
Check the source to verify that this is the current version before use.

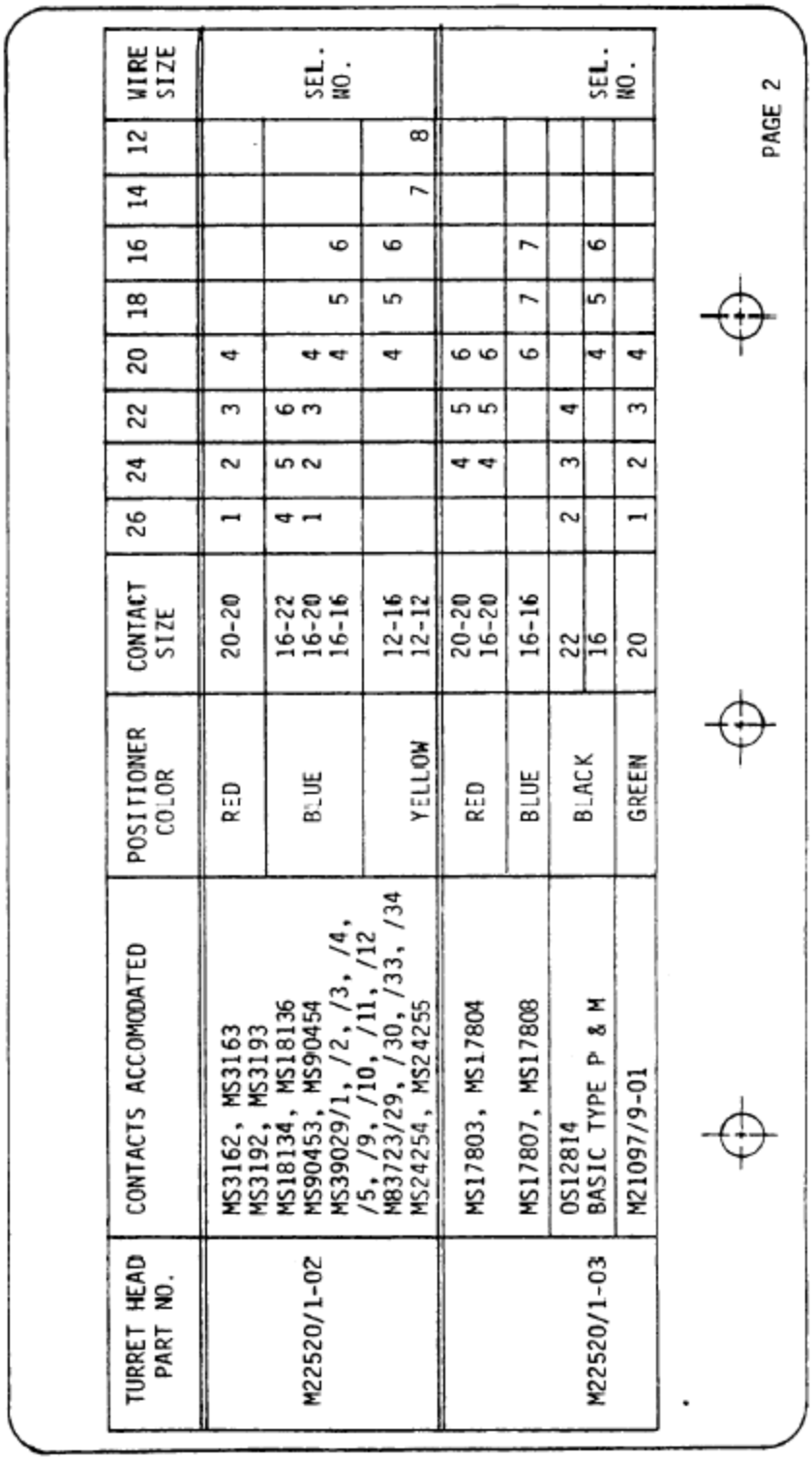


FIGURE 7 Metal case. Instruction sheet.

Source: https://assist.dla.mil -- Downloaded: 2026-10-23T08:22Z
Check the source to verify that this is the current version before use.

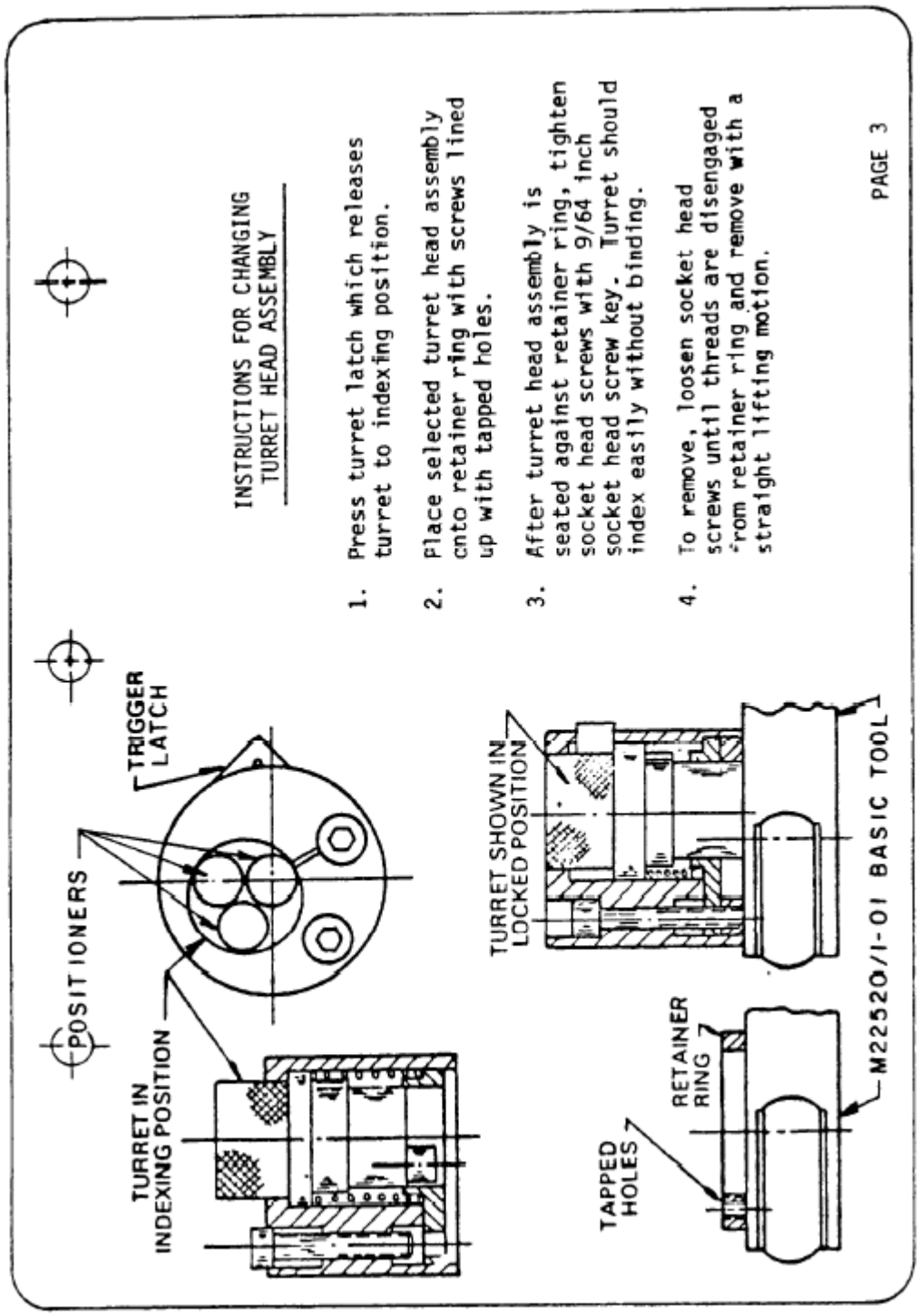


FIGURE 8 Metal case. Instruction sheet.

Source: https://assist.dla.mil -- Downloaded: 2026-10-23T08:22Z
Check the source to verify that this is the current version before use.

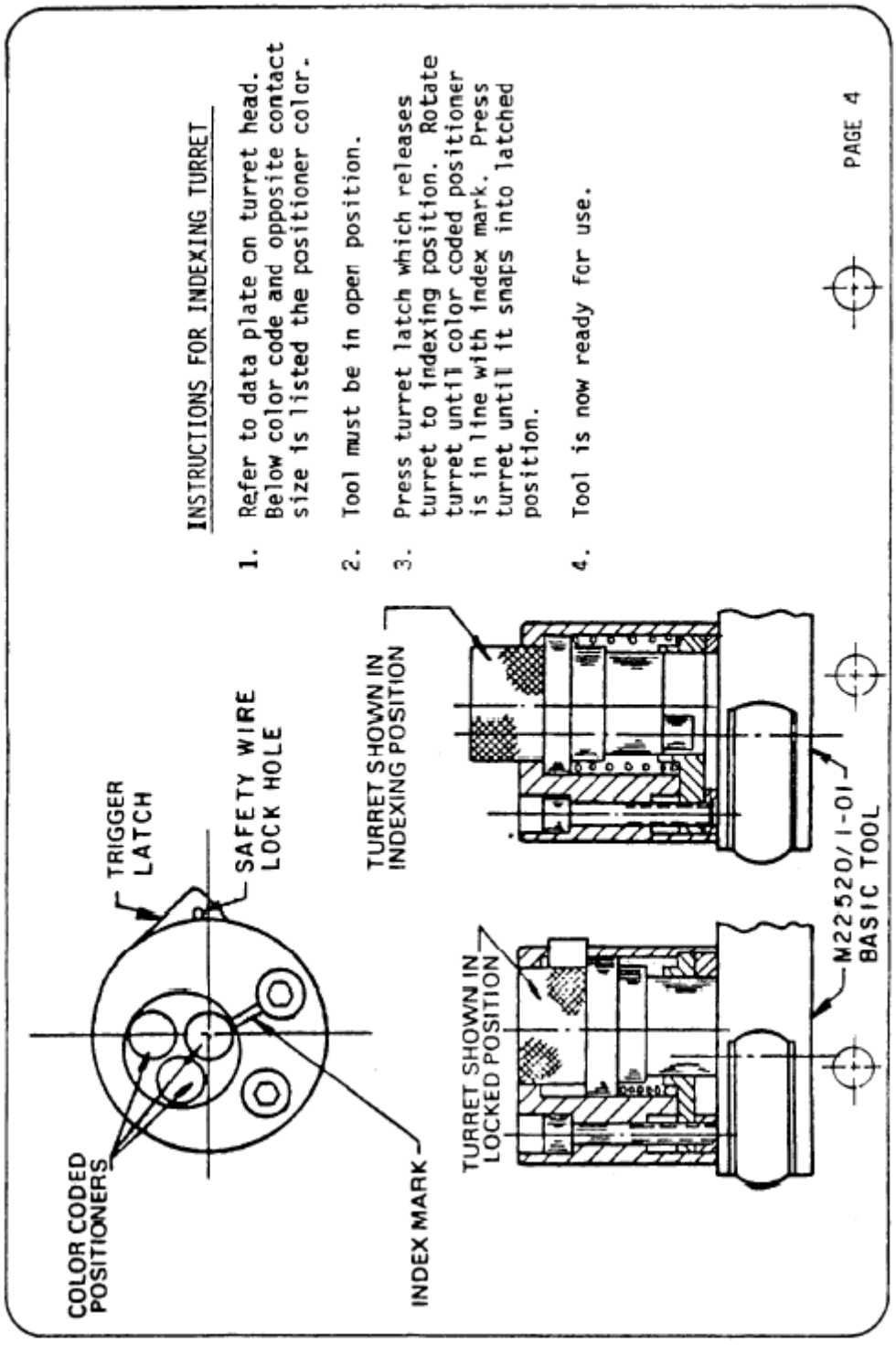


FIGURE 9 Metal case. Instruction sheet.

Source: https://assist.dla.mil -- Downloaded: 2026-10-23T08:22Z
Check the source to verify that this is the current version before use.

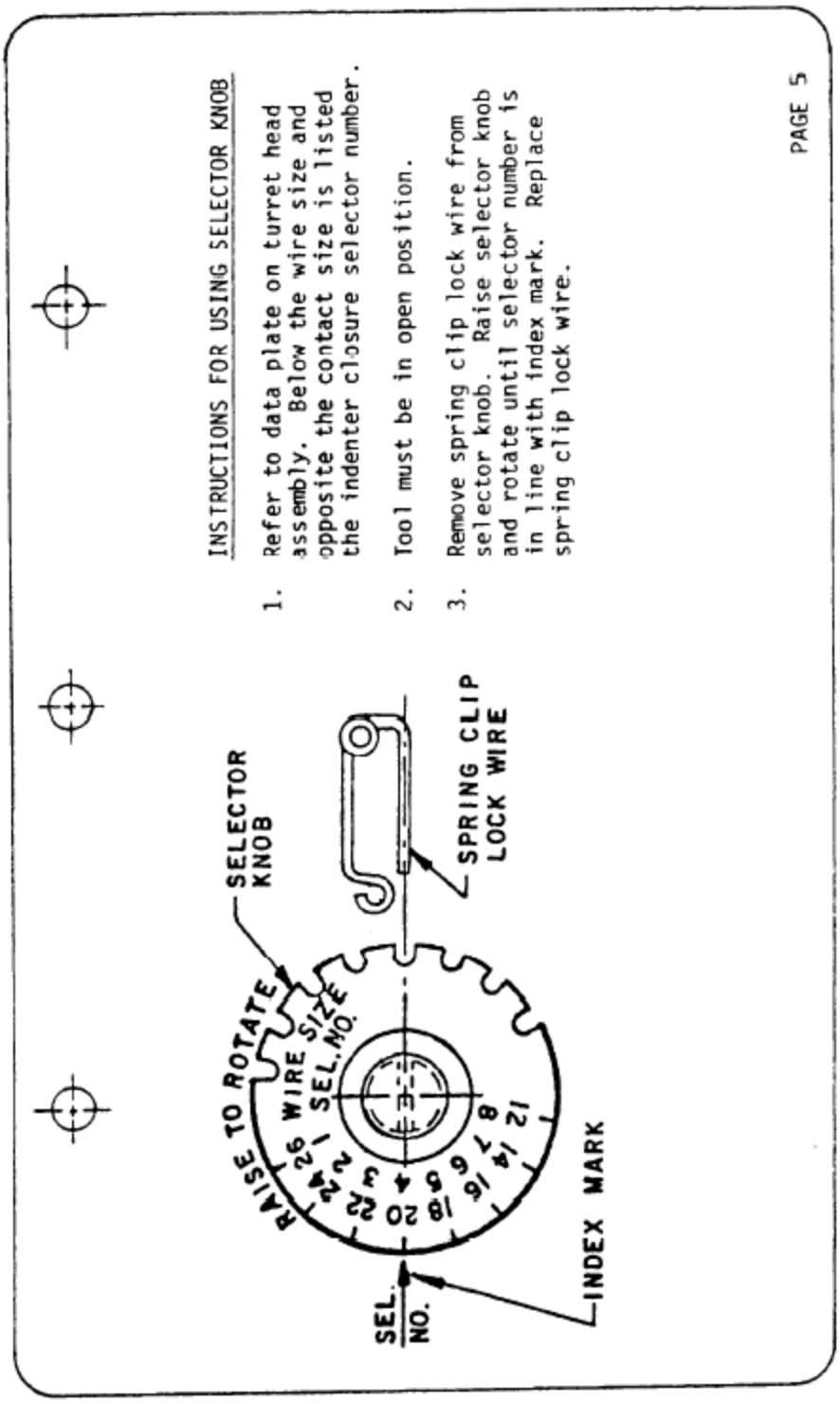


FIGURE 10 Metal case. Instruction sheet.

Source: https://assist.dla.mil -- Downloaded: 2026-10-23T08:22Z
Check the source to verify that this is the current version before use.

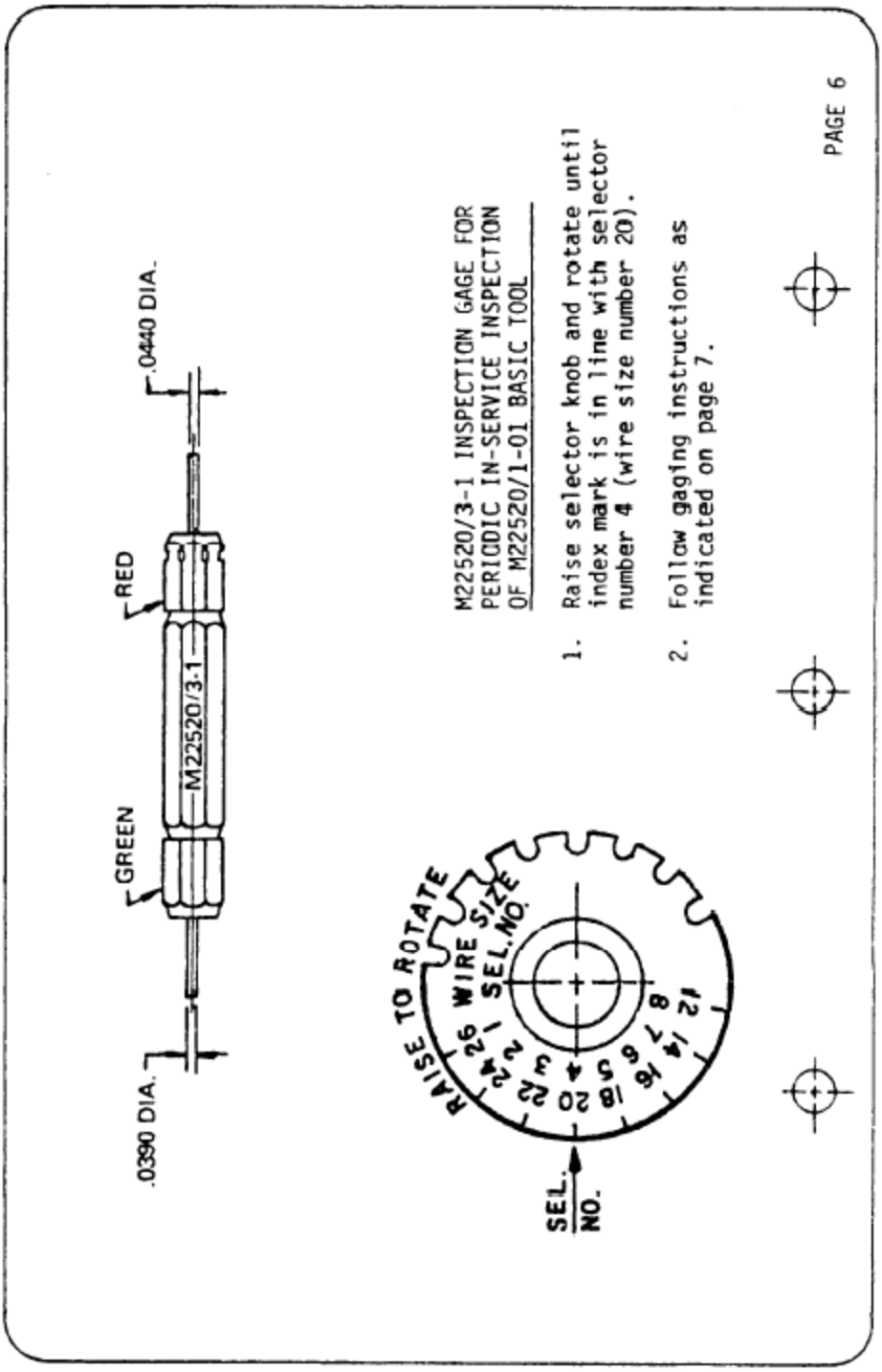


FIGURE 11 Metal case. Instruction sheet.

Source: https://assist.dla.mil -- Downloaded: 2026-10-23T08:22Z
Check the source to verify that this is the current version before use.

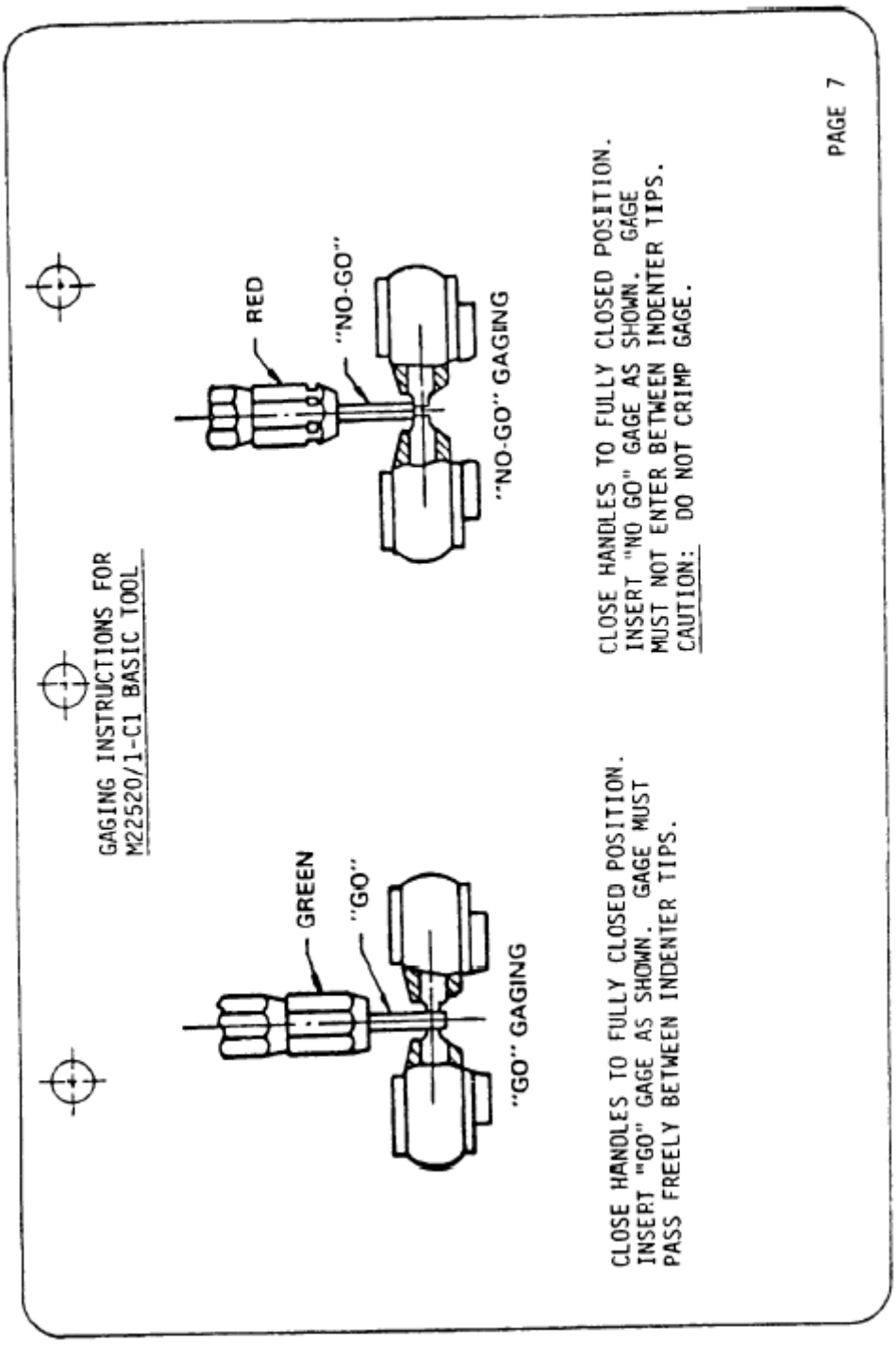
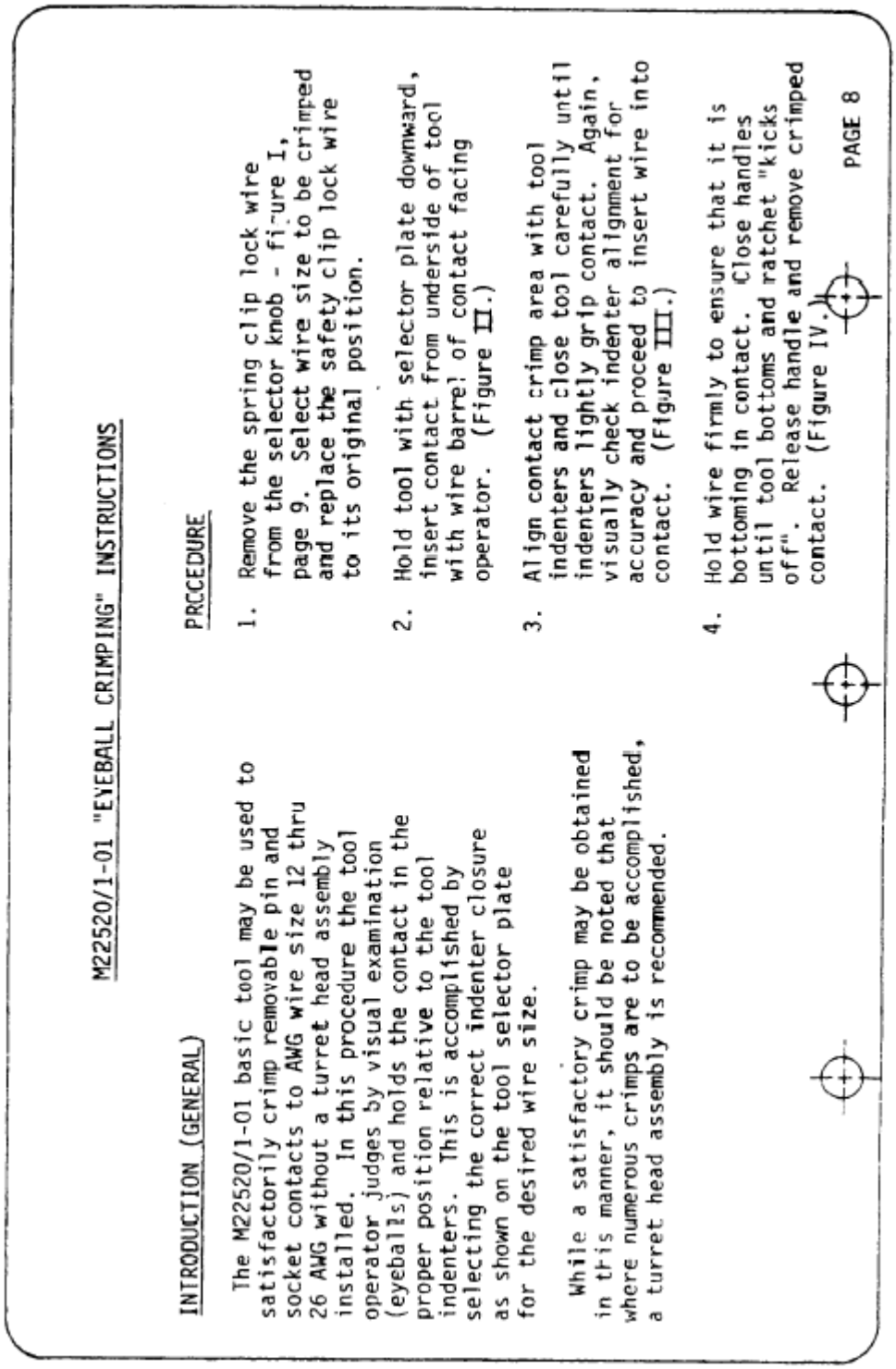


FIGURE 12 Metal case. Instruction sheet.

Source: https://assist.dla.mil -- Downloaded: 2026-10-23T08:22Z
Check the source to verify that this is the current version before use.



M22520/1-01 "EYEBALL CRIMPING" INSTRUCTIONS

INTRODUCTION (GENERAL)

The M22520/1-01 basic tool may be used to satisfactorily crimp removable pin and socket contacts to AWG wire size 12 thru 26 AWG without a turret head assembly installed. In this procedure the tool operator judges by visual examination (eyeballs) and holds the contact in the proper position relative to the tool indenters. This is accomplished by selecting the correct indenter closure as shown on the tool selector plate for the desired wire size.

While a satisfactory crimp may be obtained in this manner, it should be noted that where numerous crimps are to be accomplished, a turret head assembly is recommended.

PROCEDURE

1. Remove the spring clip lock wire from the selector knob - figure I, page 9. Select wire size to be crimped and replace the safety clip lock wire to its original position.

2. Hold tool with selector plate downward, insert contact from underside of tool with wire barrel of contact facing operator. (Figure II.)

3. Align contact crimp area with tool indenters and close tool carefully until indenters lightly grip contact. Again, visually check indenter alignment for accuracy and proceed to insert wire into contact. (Figure III.)

4. Hold wire firmly to ensure that it is bottoming in contact. Close handles until tool bottoms and ratchet "kicks off". Release handle and remove crimped contact. (Figure IV.)

PAGE 8

FIGURE 13 Metal case. Instruction sheet.

Source: https://assist.dla.mil -- Downloaded: 2026-10-23T08:22Z
Check the source to verify that this is the current version before use.

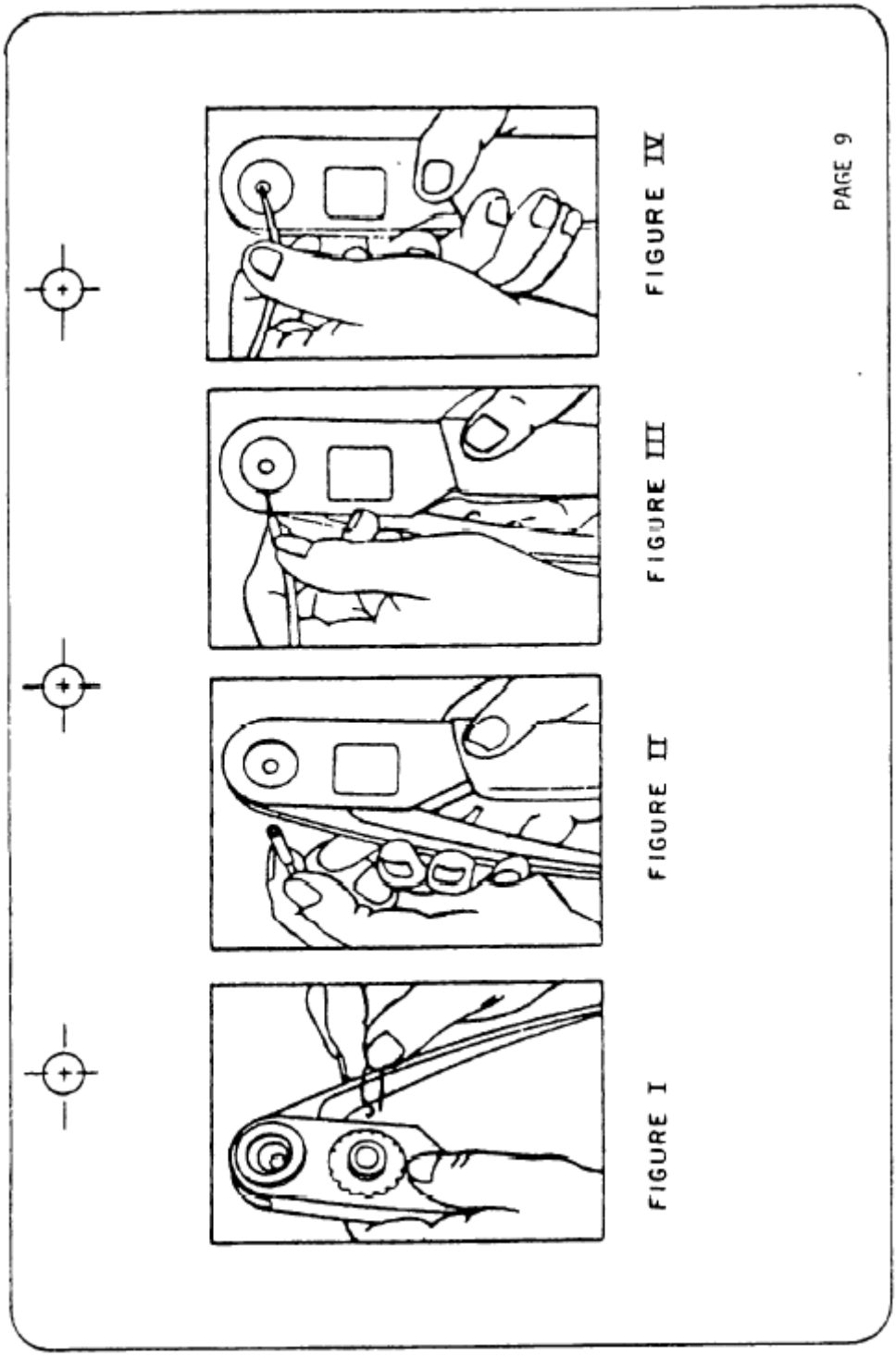


FIGURE 14 Metal case. Instruction sheet.

Source: https://assist.dla.mil -- Downloaded: 2026-10-23T08:22Z
Check the source to verify that this is the current version before use.

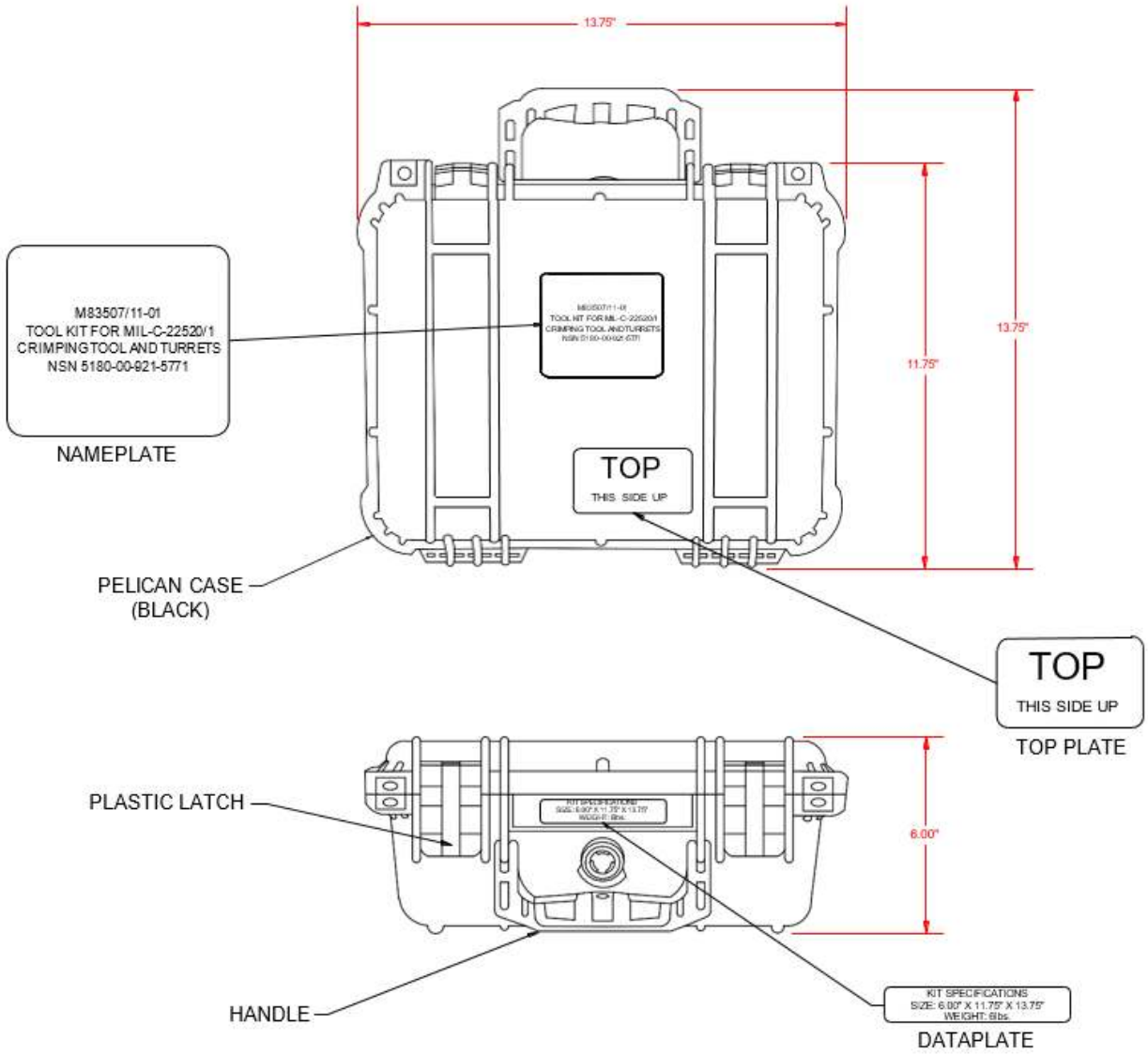


FIGURE 15 Plastic case. TOOL KIT FOR SAE-AS22520/1 CRIMPING TOOL AND TURRETS.

Source: https://assist.dla.mil -- Downloaded: 2026-10-23T08:22Z
Check the source to verify that this is the current version before use.

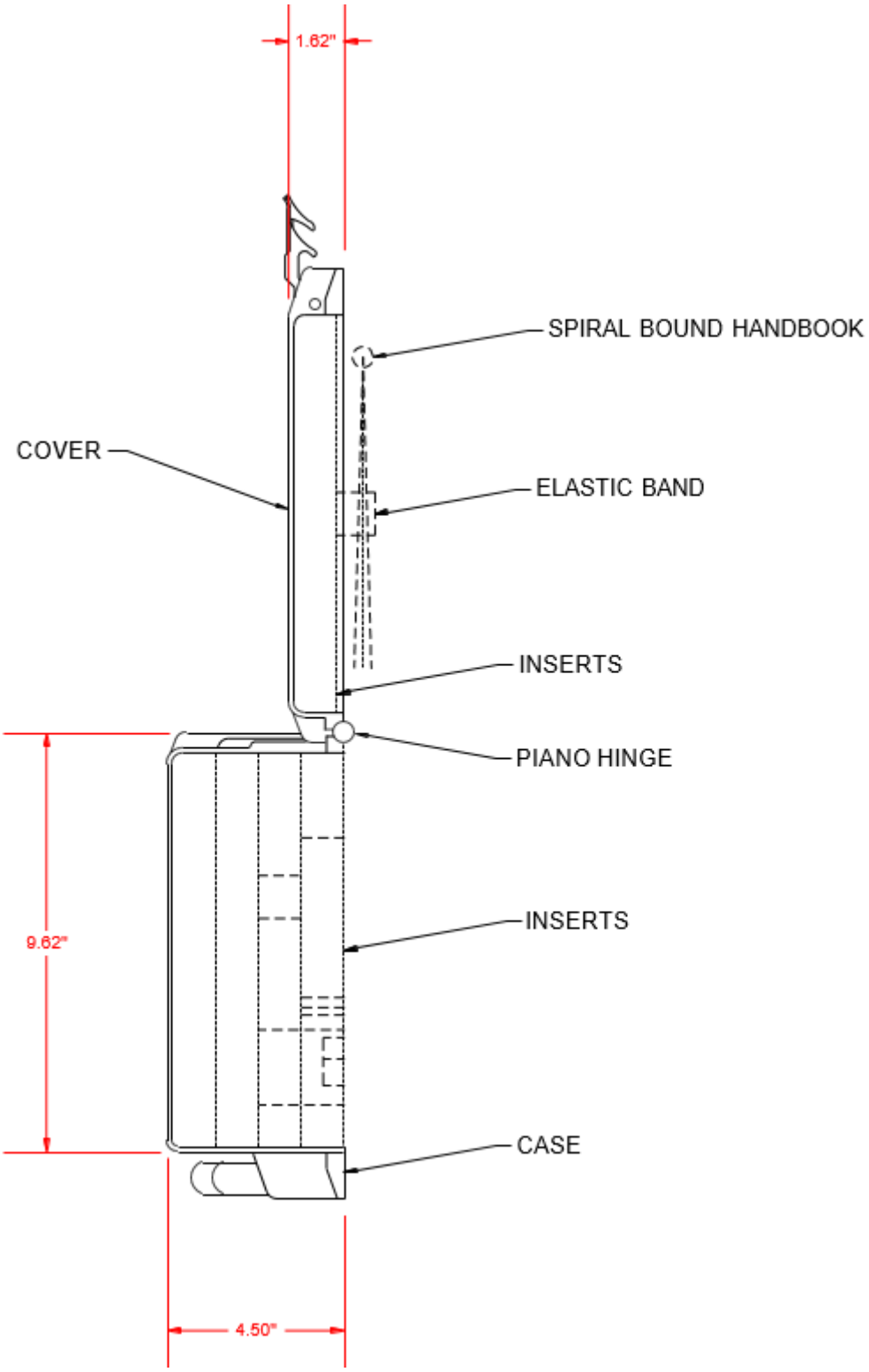


FIGURE 16 Plastic case. TOOL CASE ASSEMBLY.

Source: https://assist.dla.mil -- Downloaded: 2026-10-23T08:22Z
Check the source to verify that this is the current version before use.

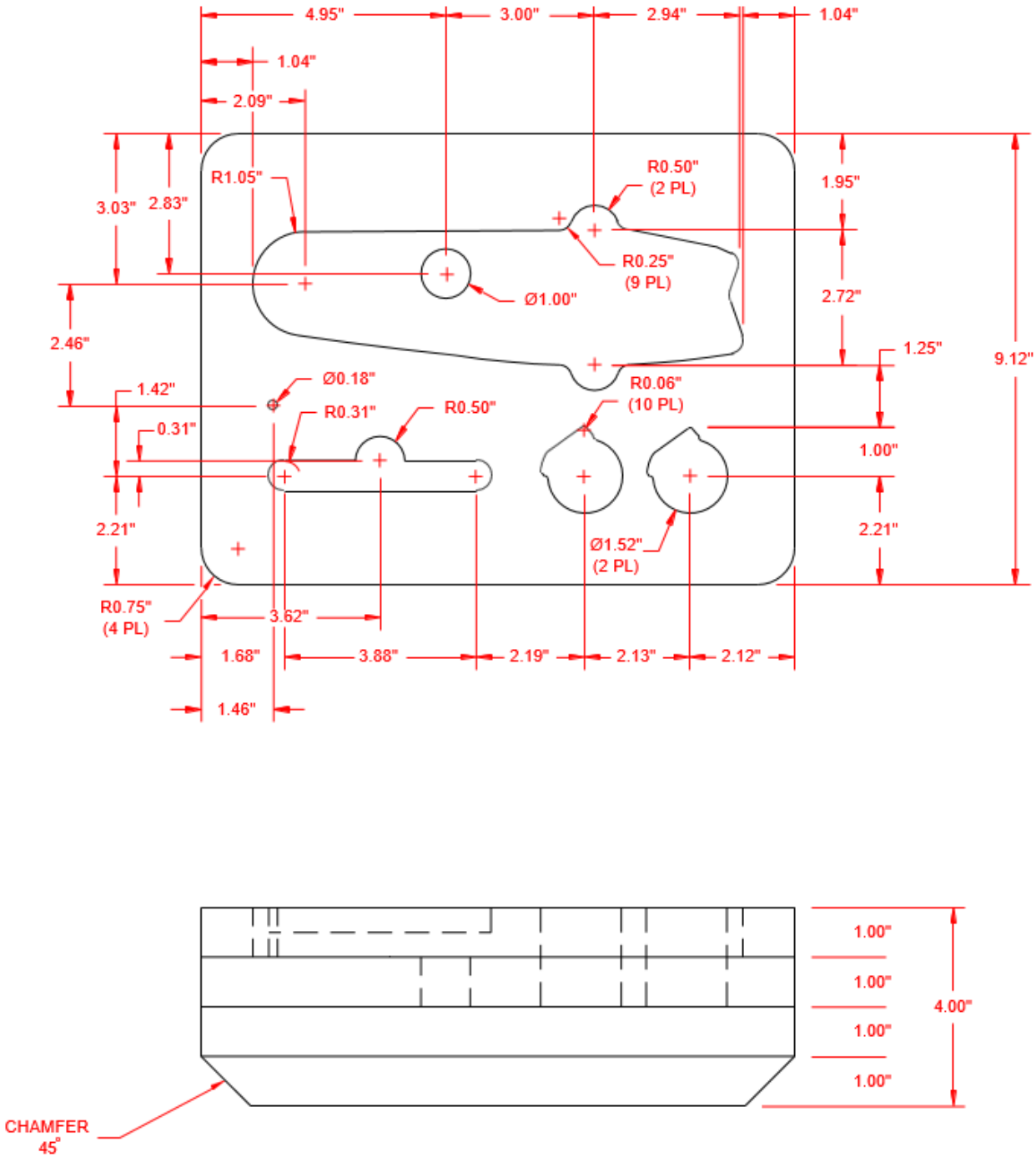


FIGURE 17 Plastic case. INSERT, BOTTOM OF CASE

Source: https://assist.dla.mil -- Downloaded: 2026-10-23T08:22Z
Check the source to verify that this is the current version before use.

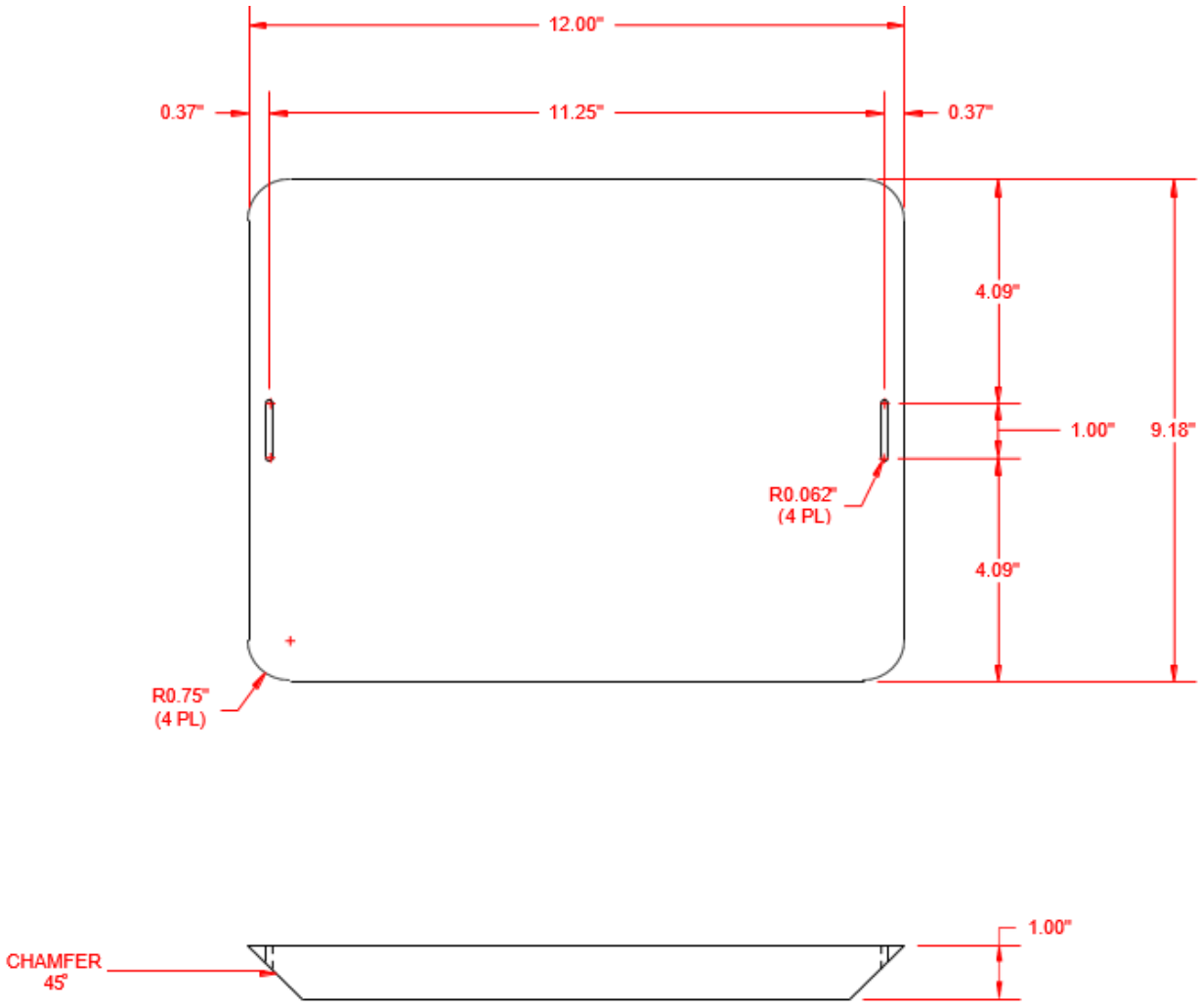


FIGURE 18 Plastic case. INSERT, TOP OF CASE

Source: https://assist.dla.mil -- Downloaded: 2026-10-23T08:22Z
Check the source to verify that this is the current version before use.

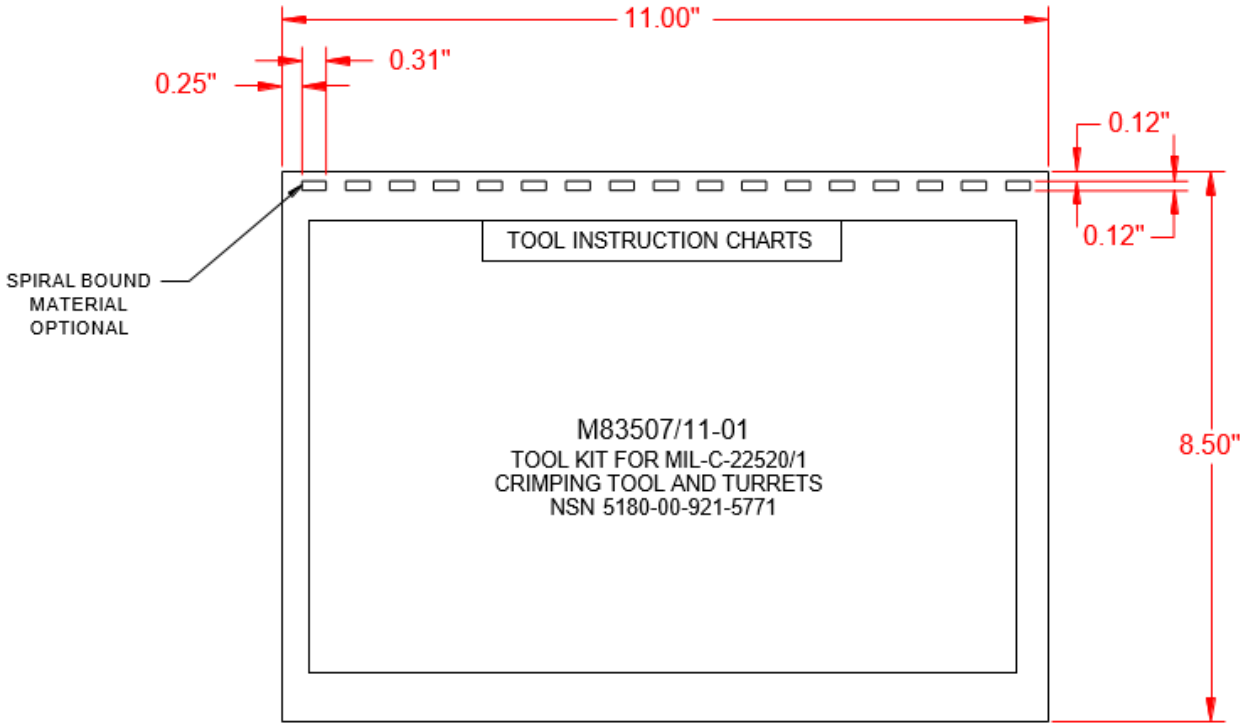


FIGURE 19 Plastic case. Instruction sheet.

Source: https://assist.dla.mil -- Downloaded: 2026-10-23T08:22Z
Check the source to verify that this is the current version before use.

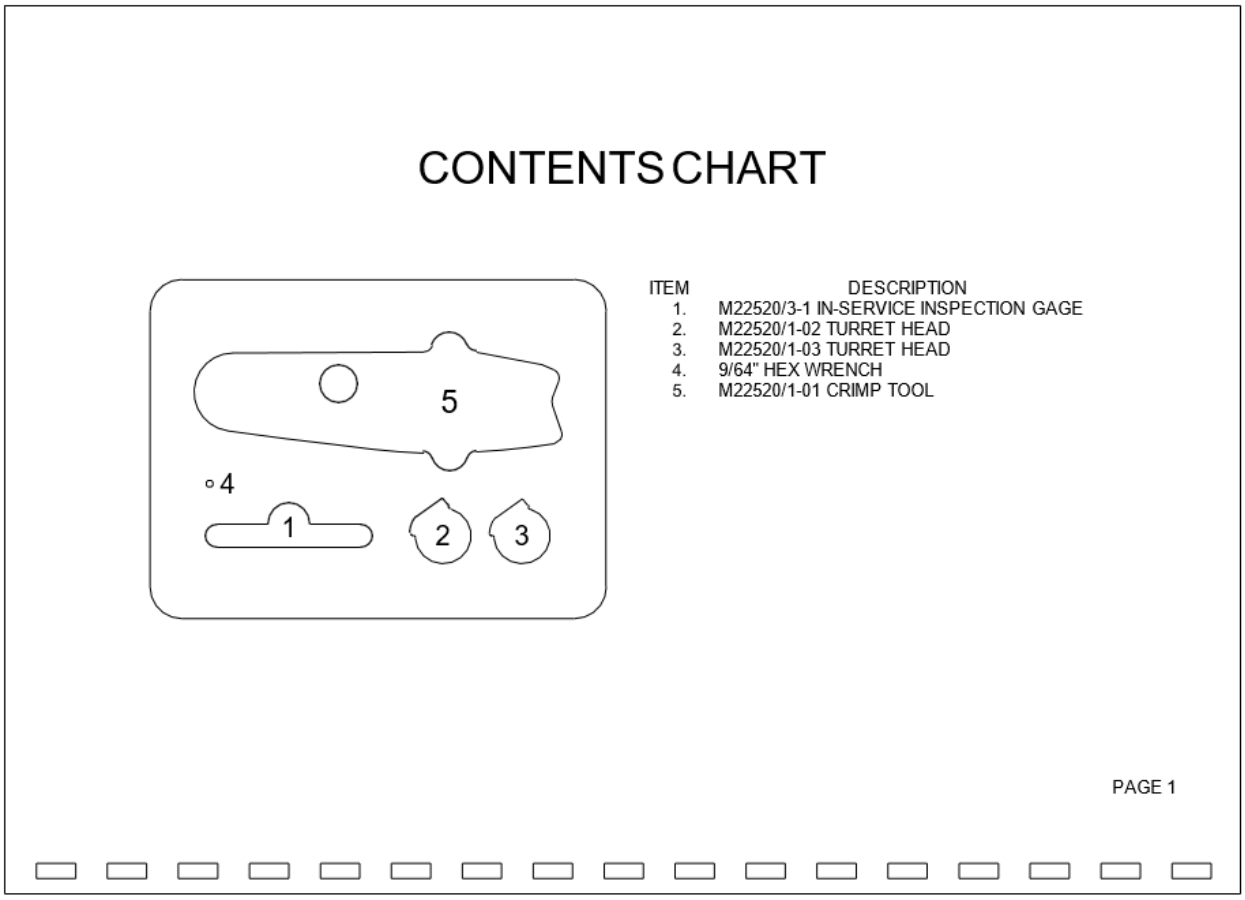
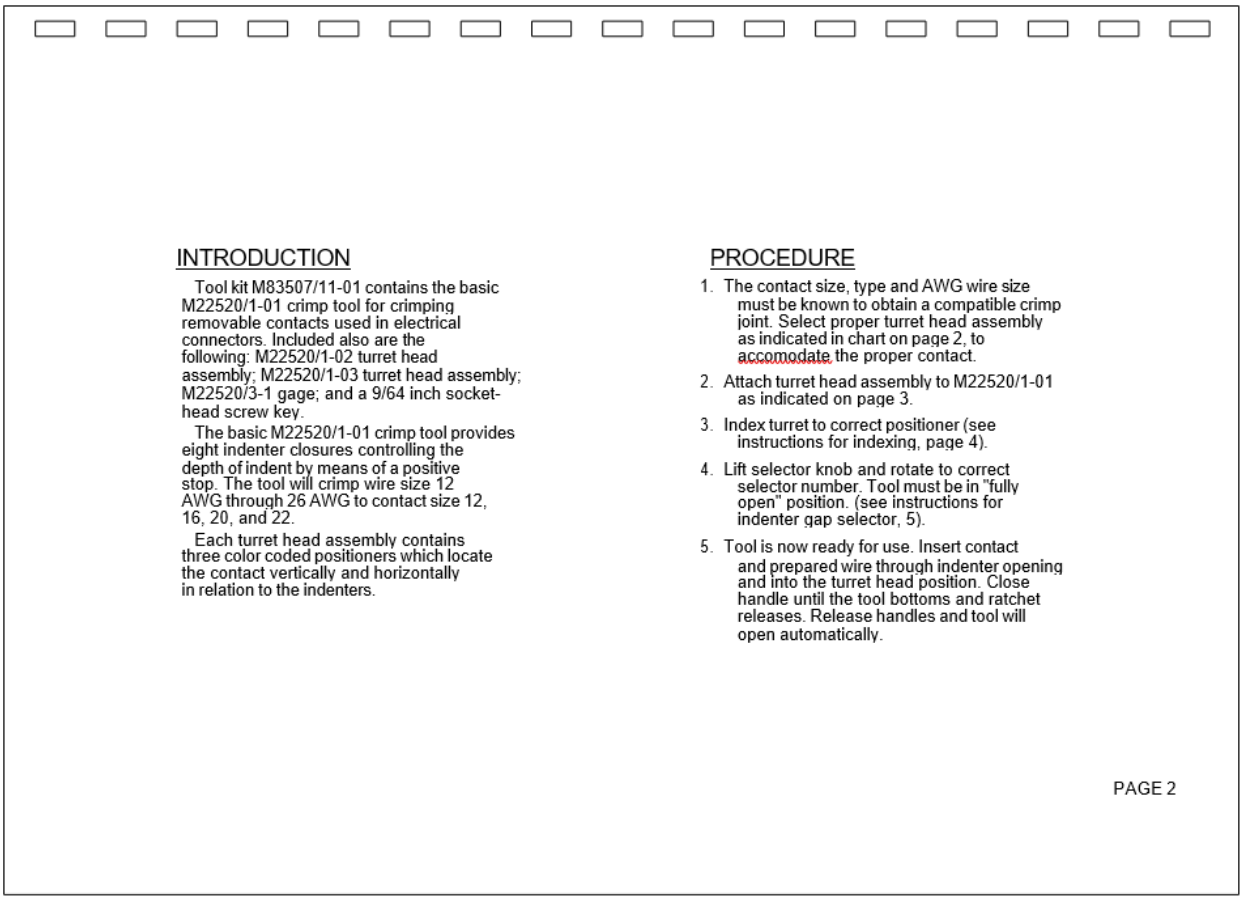


FIGURE 20 Plastic case. Instruction sheet.

Source: https://assist.dla.mil -- Downloaded: 2026-10-23T08:22Z
Check the source to verify that this is the current version before use.



INTRODUCTION

Tool kit M83507/11-01 contains the basic M22520/1-01 crimp tool for crimping removable contacts used in electrical connectors. Included also are the following: M22520/1-02 turret head assembly; M22520/1-03 turret head assembly; M22520/3-1 gage; and a 9/64 inch socket-head screw key.

The basic M22520/1-01 crimp tool provides eight indenter closures controlling the depth of indent by means of a positive stop. The tool will crimp wire size 12 AWG through 26 AWG to contact size 12, 16, 20, and 22.

Each turret head assembly contains three color coded positioners which locate the contact vertically and horizontally in relation to the indenters.

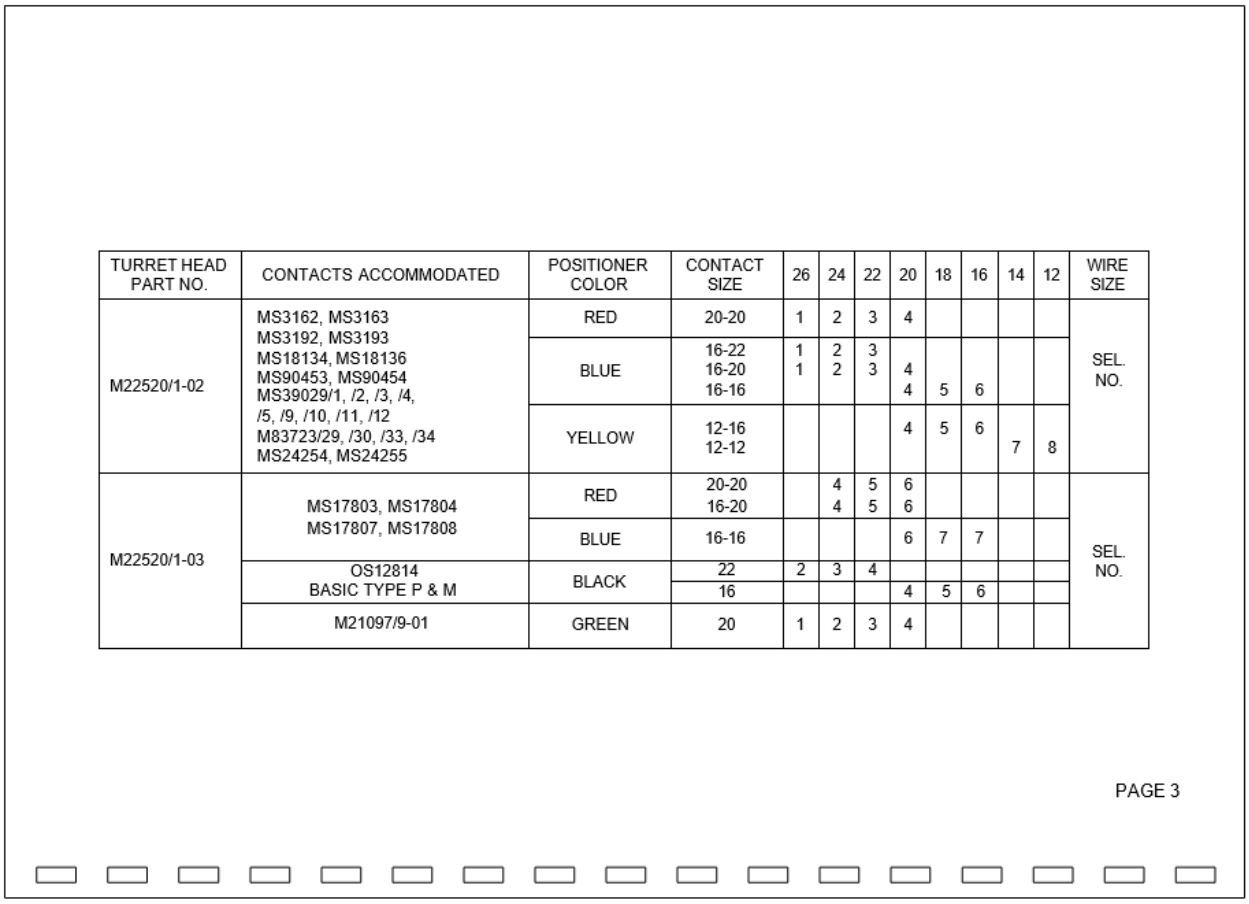
PROCEDURE

1. The contact size, type and AWG wire size must be known to obtain a compatible crimp joint. Select proper turret head assembly as indicated in chart on page 2, to accomodate the proper contact.
2. Attach turret head assembly to M22520/1-01 as indicated on page 3.
3. Index turret to correct positioner (see instructions for indexing, page 4).
4. Lift selector knob and rotate to correct selector number. Tool must be in "fully open" position. (see instructions for indenter gap selector, 5).
5. Tool is now ready for use. Insert contact and prepared wire through indenter opening and into the turret head position. Close handle until the tool bottoms and ratchet releases. Release handles and tool will open automatically.

PAGE 2

FIGURE 21 Plastic case. Instruction sheet.

Source: https://assist.dla.mil -- Downloaded: 2026-10-23T08:22Z
Check the source to verify that this is the current version before use.

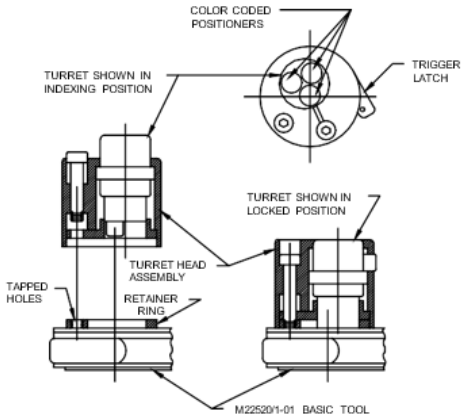


TURRET HEAD PART NO.	CONTACTS ACCOMMODATED	POSITIONER COLOR	CONTACT SIZE	26	24	22	20	18	16	14	12	WIRE SIZE
M22520/1-02	MS3162, MS3163 MS3192, MS3193 MS18134, MS18136 MS90453, MS90454 MS39029/1, /2, /3, /4, /5, /9, /10, /11, /12 M83723/29, /30, /33, /34 MS24254, MS24255	RED	20-20	1	2	3	4					SEL. NO.
		BLUE	16-22 16-20 16-16	1 1	2 2	3 3	4 4	5	6			
		YELLOW	12-16 12-12				4	5	6	7	8	
M22520/1-03	MS17803, MS17804 MS17807, MS17808	RED	20-20 16-20		4 4	5 5	6 6					SEL. NO.
		BLUE	16-16				6	7	7			
	OS12814 BASIC TYPE P & M	BLACK	22	2	3	4						
			16				4	5	6			
	M21097/9-01	GREEN	20	1	2	3	4					

PAGE 3

FIGURE 22 Plastic case. Instruction sheet.

Source: https://assist.dla.mil -- Downloaded: 2026-10-23T08:22Z
Check the source to verify that this is the current version before use.



INSTRUCTIONS FOR CHANGING TURRET HEAD ASSEMBLY

1. Press turret latch which releases turret to indexing position.
2. Place selected turret head assembly onto retainer ring with screws lined up with tapped holes.
3. After turret head assembly is seated against retainer ring, tighten socket head screws with 9/64 inch socket head screw key. Turret should index easily without binding.
4. To remove, loosen socket head screws until threads are disengaged from retainer ring and remove with a straight lifting motion.

PAGE 4

FIGURE 23 Plastic case. Instruction sheet.

Source: https://assist.dla.mil -- Downloaded: 2026-10-23T08:22Z
Check the source to verify that this is the current version before use.

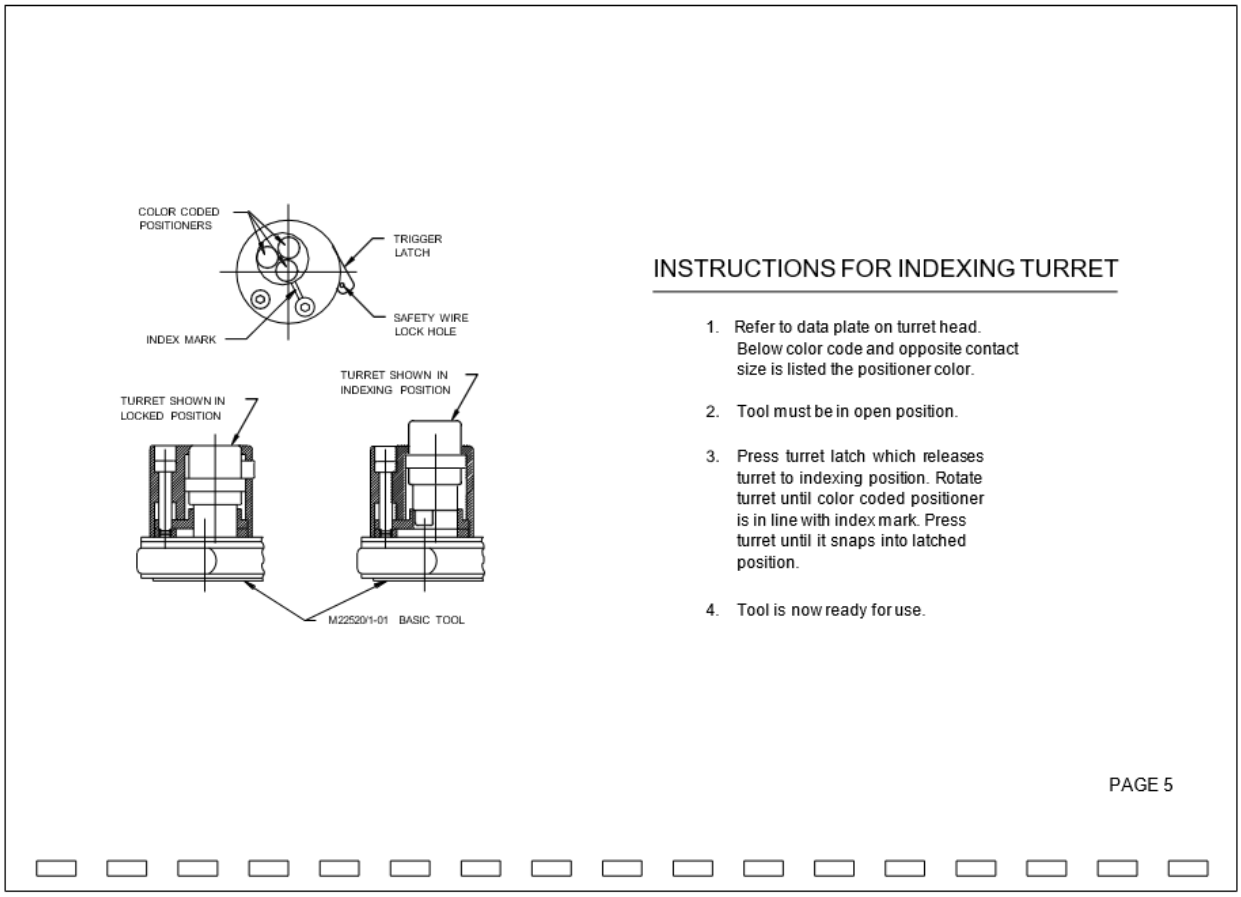


FIGURE 24 Plastic case. Instruction sheet.

Source: https://assist.dla.mil -- Downloaded: 2026-10-23T08:22Z
Check the source to verify that this is the current version before use.

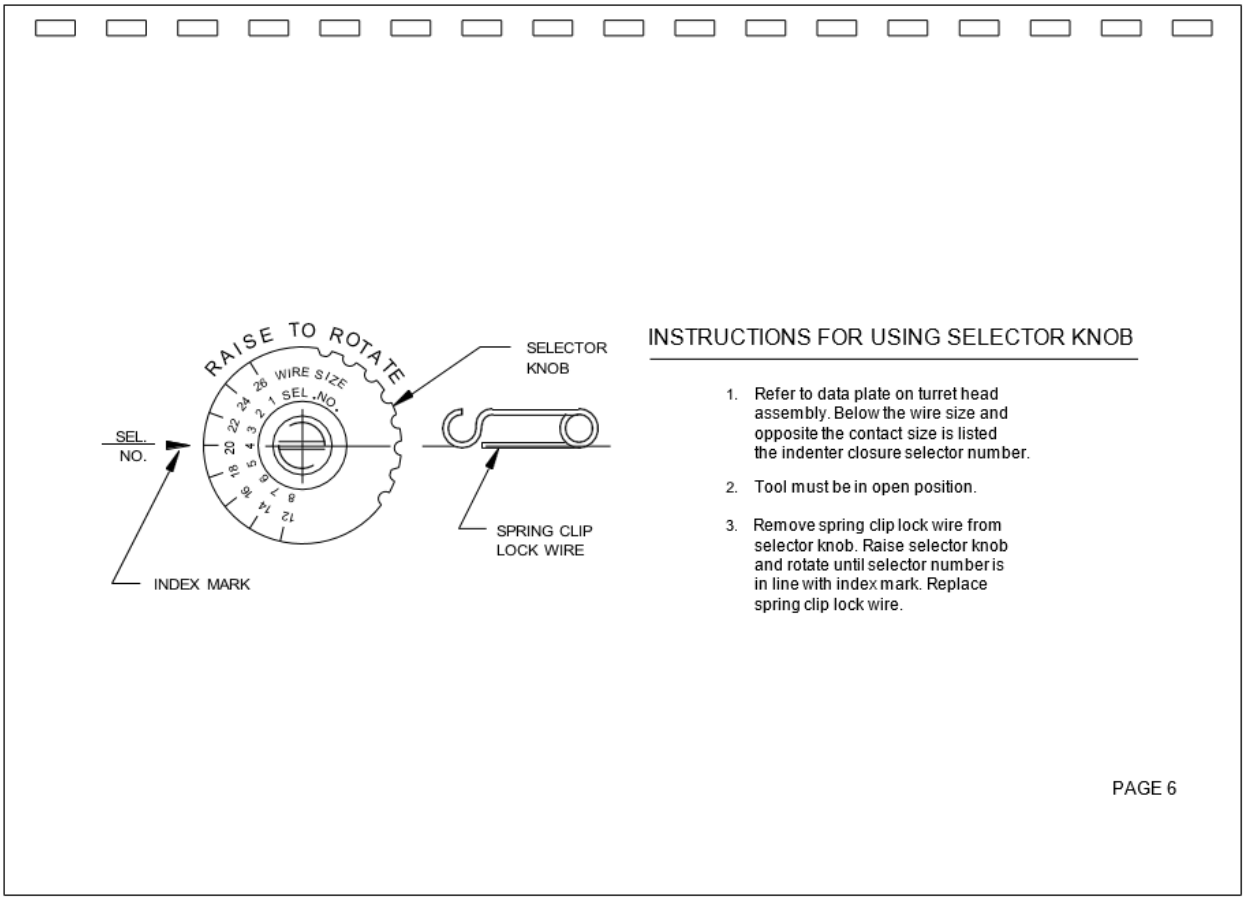


FIGURE 25 Plastic case. Instruction sheet.

Source: https://assist.dla.mil -- Downloaded: 2026-10-23T08:22Z
Check the source to verify that this is the current version before use.

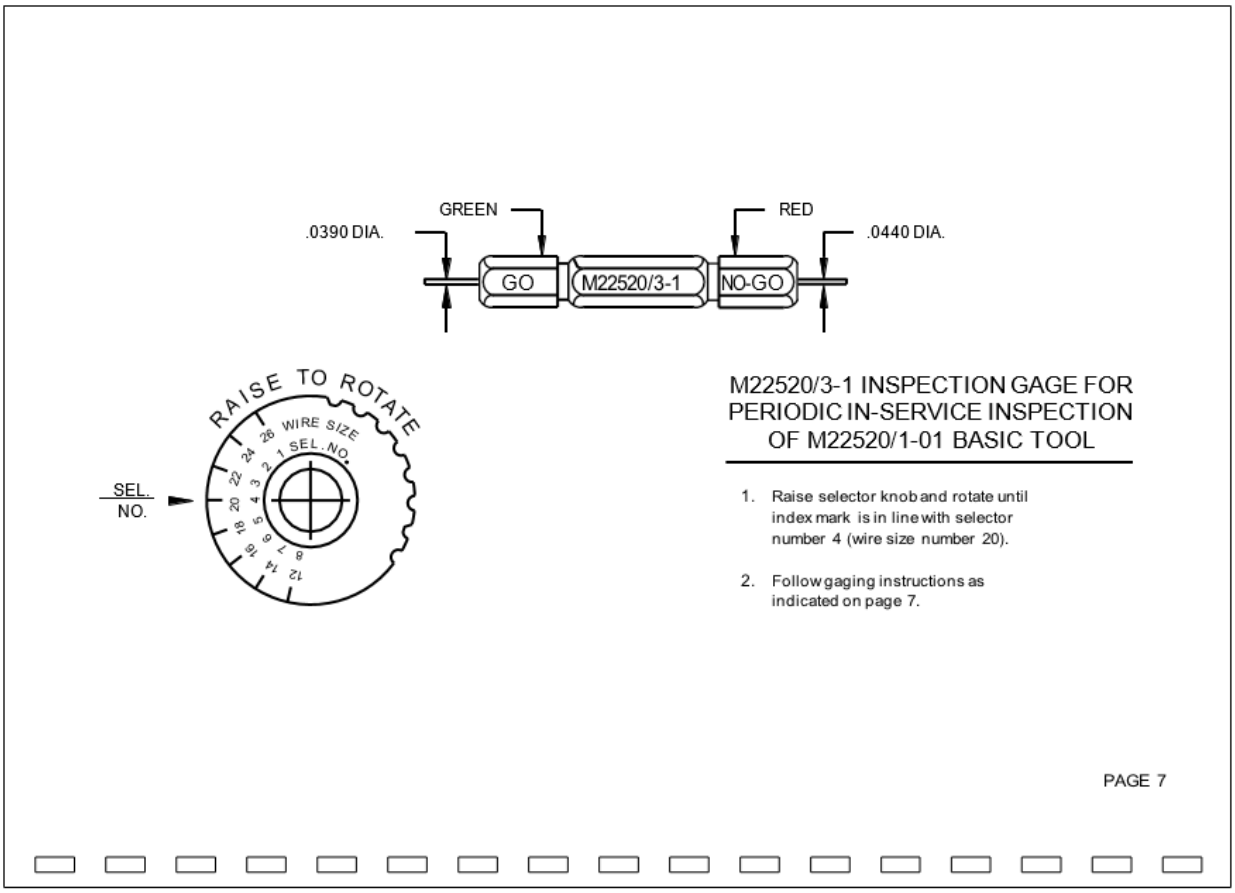


FIGURE 26 Plastic case. Instruction sheet.

Source: https://assist.dla.mil -- Downloaded: 2026-10-23T08:22Z
Check the source to verify that this is the current version before use.

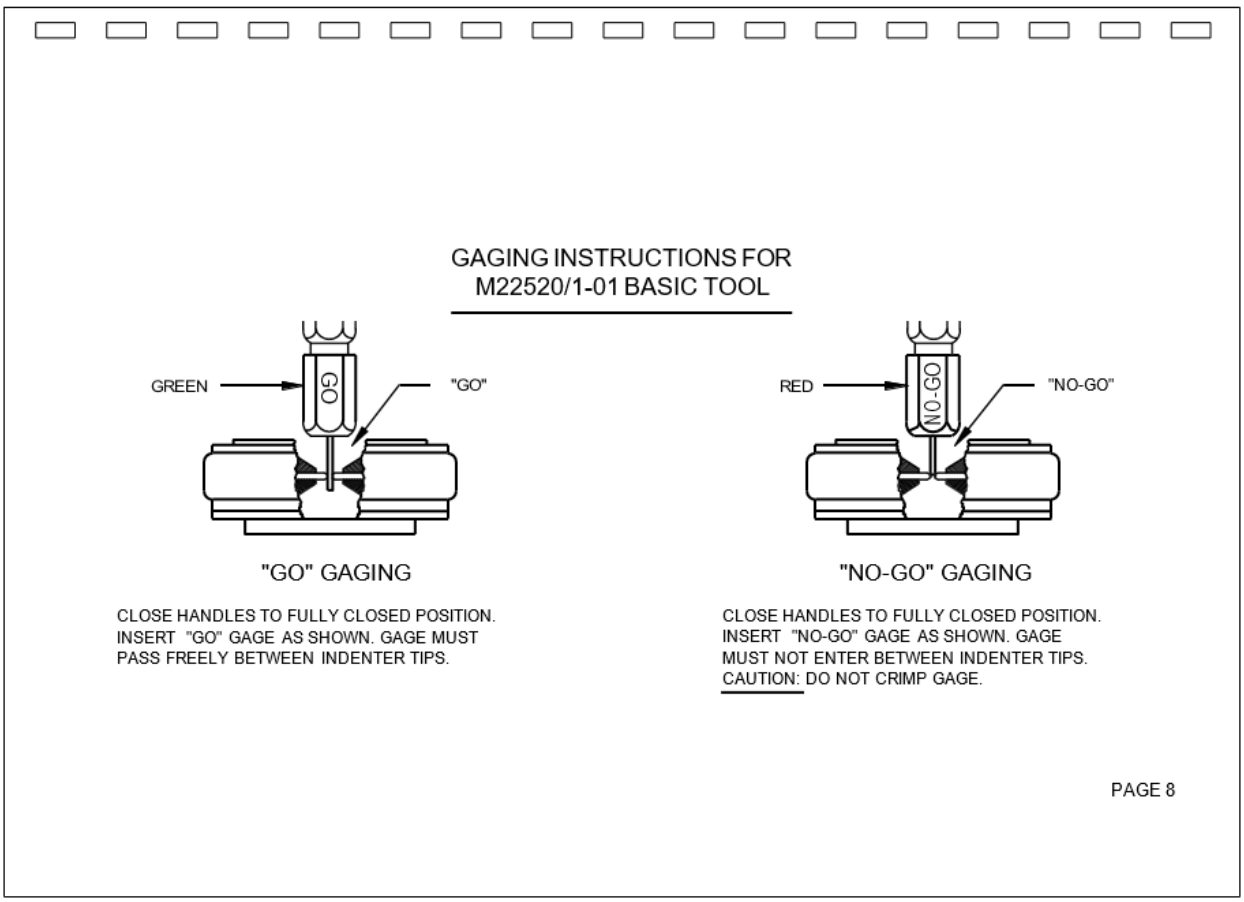
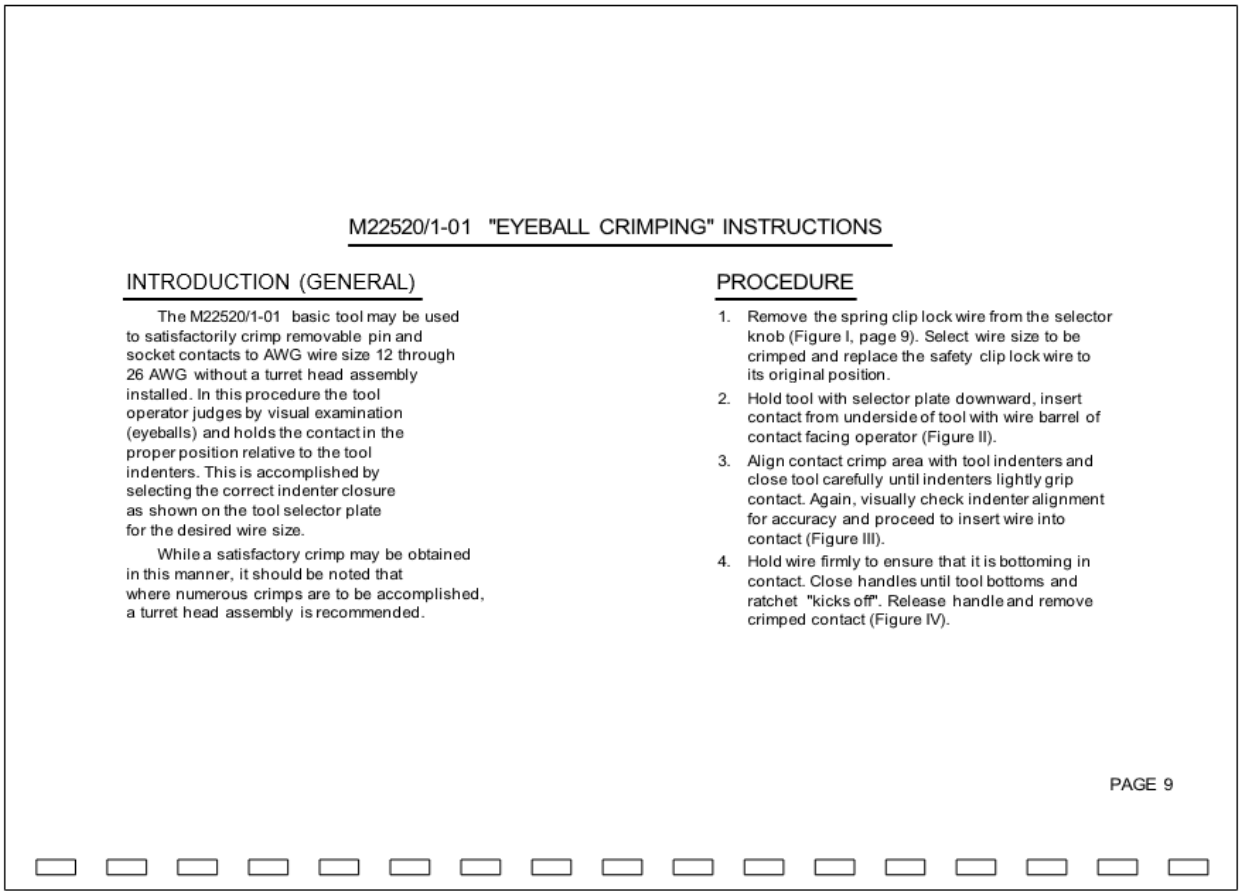


FIGURE 27 Plastic case. Instruction sheet.

Source: https://assist.dla.mil -- Downloaded: 2026-10-23T08:22Z
Check the source to verify that this is the current version before use.



M22520/1-01 "EYEBALL CRIMPING" INSTRUCTIONS

INTRODUCTION (GENERAL)

The M22520/1-01 basic tool may be used to satisfactorily crimp removable pin and socket contacts to AWG wire size 12 through 26 AWG without a turret head assembly installed. In this procedure the tool operator judges by visual examination (eyeballs) and holds the contact in the proper position relative to the tool indenters. This is accomplished by selecting the correct indenter closure as shown on the tool selector plate for the desired wire size.

While a satisfactory crimp may be obtained in this manner, it should be noted that where numerous crimps are to be accomplished, a turret head assembly is recommended.

PROCEDURE

1. Remove the spring clip lock wire from the selector knob (Figure I, page 9). Select wire size to be crimped and replace the safety clip lock wire to its original position.
2. Hold tool with selector plate downward, insert contact from underside of tool with wire barrel of contact facing operator (Figure II).
3. Align contact crimp area with tool indenters and close tool carefully until indenters lightly grip contact. Again, visually check indenter alignment for accuracy and proceed to insert wire into contact (Figure III).
4. Hold wire firmly to ensure that it is bottoming in contact. Close handles until tool bottoms and ratchet "kicks off". Release handle and remove crimped contact (Figure IV).

PAGE 9

FIGURE 28 Plastic case. Instruction sheet.

Source: https://assist.dla.mil -- Downloaded: 2026-10-23T08:22Z
Check the source to verify that this is the current version before use.

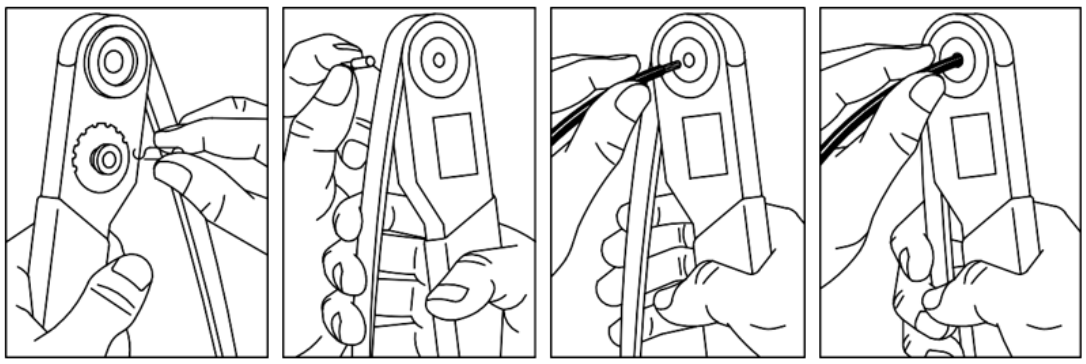


FIGURE I FIGURE II FIGURE III FIGURE IV

PAGE 10

FIGURE 29 Plastic case. Instruction sheet.

REQUIREMENTS:

Materials:

Metal case and cover (category 3):

Sheet metal, 21 gage in accordance with QQ-S-698; gray baked enamel on all metal parts.

Pelican plastic case

Instruction sheet and contents chart: White vinyl 0.020 inch (0.51 rm,) thick. Insert and retainer pad: fire retardant foamed polythylene in accordance with A-A-59135 or A-A-59136; color is optional.

Clasps, draw pull type with provision for locking device, handles, and hinges: Suitably plated to resist corrosion.

Nameplate: In accordance with MIL-DTL-19834.

Binder (metal case): Looseleaf, three ring, 0.437 inch (11.10 nm) :t0.062 inch (1.57 rm,) I.D., 2.75 inches

(69.9 mm) center distance between rings, securely riveted to inside of cover.

Instruction sheets and contents chart: Lettering shall be legible and protected with a transparent coating. Contents chart (centered on·inside of case cover) shall have pressure sensitive adhesive backing and show the insert cavity configurations and the respective tool part numbers; a list of NSN's will also be supplied.

Source: https://assist.dla.mil -- Downloaded: 2026-10-23T08:22Z
Check the source to verify that this is the current version before use.

Part number:
M83507/11-01 Tool kit for SAE-AS22520/1 crimping tool and turrets.
M83507/11-02 Tool case assembly.

Changes from previous issue. Marginal notations are not used in this revision to identify changes with respect to the previous issue due to the extent of the changes.

Referenced documents. In addition to MIL-DTL-83507, this document references the following:

A-A-59135
A-A-59136
MIL-DTL-19834
QQ-S-698
SAE-AS22520/1

CONCLUDING MATERIAL

Custodians:
Navy – AS

Preparing activity:
DLA - CC

Review activities:
Navy – EC, MC
DLA - GS

(Project 5180-2025-003)

NOTE: The activities listed above were interested in this document as of the date of this document. Since organizations and responsibilities can change, you should verify the currency of the information above using the ASSIST Online database at https://assist.dla.mil.

Source: https://assist.dla.mil -- Downloaded: 2026-10-23T08:22Z
Check the source to verify that this is the current version before use.